- Rebuilt from my own research project, Play to Learn in VR (P01.01), scaled to a study one student can finish in this course: one coder, about three hours of coding, 150 items.
- Everything under "The example" is what a student would submit: a title, then the six headed sections and the timeline the template asks for.
- The rubric map at the end shows which grading row each part earns and why. Your topic and your data source are your own choice.

## The example

**Learning Talk and VR Tone: Affordances and Constraints in Tweets About Playing in Virtual Reality**

### *Research questions and hypotheses*

H1: Tweets about playing in VR that refer to learning are more often positive toward VR than those that do not. RQ1: Which affordances and constraints of VR appear most often in tweets that refer to learning?

### *Theoretical framework*

Affordance theory holds that people see a tool in terms of what it lets them do, its affordances, and what stops them, its constraints (Gibson, 1979). A writer who brings learning into a tweet about VR is describing VR as a place that affords learning, so the theory predicts that tweets referring to learning will speak of VR more favorably (H1), and it sorts what they name into affordances and constraints (RQ1).

### *The gap*

The largest study of VR-in-education talk scored 17 million tweets with a word list, found it mostly positive, and never separated play or compared learning tweets with the rest (Lampropoulos et al., 2022). Studies of how people feel about learning in VR ask participants after a designed lesson (Allcoat & von Mühlenen, 2018; Makransky & Lilleholt, 2018). Neither hand-codes what users volunteer about playing in VR.

### *Method overview*

The unit of analysis is one tweet, not a retweet, posted in 2021 and mentioning VR with play, from an archive of 100,363 such tweets collected through the Twitter Academic Research API; I record the collection date and drop sensitive-flagged tweets. I will draw 75 tweets with a learning word and 75 without, 150 in all, the 75-seconds-each sizing row, about three hours of coding. I will code each tweet for whether it refers to learning, its tone

toward VR, the affordance it names, and the constraint it names, and record the like count as a looked-up number. I code alone and will recode 10 percent after a gap to check my own consistency.

### *Expected contribution*

Teachers and designers weighing VR play for a class learn which benefits users volunteer and which problems they name, in users' own words.

### *Timeline*

| Due | Assignment | Finished by then |
|---|---|---|
| 10-16 | Definitions Practice | Four variables defined. |
| 10-23 | Codebook and Qualitative Memo | Rules and examples from 30 tweets. |
| 11-06 | Sampling Plan and Pilot | 150 tweets drawn, 20 coded and timed. |
| 11-13 | Data Wrangling in R | Pilot codes cleaned. |
| 11-20 | Describing Data in R | All 150 coded and counted by learning reference. |
| 12-04 | Inferencing Data in R | Chi-square test of H1. |
| 12-18 | White Paper | Introduction from this prospectus, plus method, results, and discussion. |

### Rubric map

Points are the assignments page's own split: four each for questions, theory, gap, and method; three each for title, contribution, and timeline.

| Rubric row | Points | Where the example earns it | Why it earns the full row |
|---|---|---|---|
| Title | 3 | The title line | It names two things that will be coded or compared (learning talk, tone toward VR) and the two categories counted (affordances, constraints), so a reader can guess the variables. It could not be the title of another study on VR. |
| Research questions or hypotheses | 4 | H1 and RQ1 | Two items, under the maximum of three. Both are answerable from reading tweets, with no one asked anything. H1 links two coded things, learning reference and tone, in a stated direction, as a tendency ("more often"), and claims an association rather than a cause. |

| Rubric row | Points | Where the example earns it | Why it earns the full row |
|---|---|---|---|
| Theoretical framework | 4 | The two sentences on affordance theory | It names the lens, says what the lens predicts a writer will do, and uses the lens twice more: it supplies H1's direction and the two categories RQ1 counts. The theory is not named and dropped. |
| The gap | 4 | The three sentences with four citations | It names the closest studies read, says what each did (word-list scoring; asking participants after a lesson), and names what they left open. That open question is the one H1 and RQ1 answer. No claim that nobody has studied this. |
| Method overview | 4 | The four method sentences | All four required parts are present: one unit of analysis stated once, a source with a boundary (platform, year, query, archive size, collection date, sensitive-flag exclusion), an amount tied to a named row of the sizing table, and how the key ideas are measured. At least one variable is coded by the student, with the like count as a looked-up number beside it. One coder, with a recode plan instead of a second coder. |
| Expected contribution | 3 | The one contribution sentence | It names who learns something (teachers and designers) and what they learn, and it claims only what coded tweets can show: what users wrote, not whether VR improves learning. |
| Timeline | 3 | The seven timeline rows | One row per remaining assignment, each with its due date and a result that could be checked off, in assignment order rather than by week. |

Length: the example runs about 400 words from the title through the timeline, inside the 250-to-400-word range, so the whole study fits on one page.